Chapter 9

Problems based on ages

Age is defined as a period of time that a person has lived or a thing has existed. Age is measured in **months, years, decades** and so on.
Problem based on ages generally consists of information of ages of two or more persons and a relation between their ages in **present/future/past**.
Using the information, it is asked to calculate the ages of one or more persons in present/future/past.

9.1 Important Rules for Problem Based on Ages

Rule I

If ratio of present ages of A and B is $x : y$ and n year ago, the ratio of their ages was $p : q$, then

$$\frac{x-n}{y-n} = \frac{p}{q}$$

Ex. 1 A father is twice as old as his son. 20 years back, he was twelve times as old as his son. What is the present age of father?

Sol. Let present age son be$k\ yr$, the present age of father will be $2k\ yr$.

Then $x = 2k$ and $y = k$

Also, 20 years before, the ratio of their ages was $12 : 1$

So, $$\frac{p}{q} = \frac{12}{1}$$

Then, $$\frac{x-n}{y-n} = \frac{p}{q} \implies \frac{2k-20}{k-20} = \frac{12}{1}$$

$$\implies \quad 2k-20 = 12(k-20) \implies 2k-20 = 12k-240$$

$$\implies \quad 12k-2k = 240-20 \implies 10k = 220$$

$$\therefore \qquad k = \frac{220}{10} = 22$$

$$\therefore \quad \text{Fathers present age} = 2 \times 22 = 44 \ yr$$

Rule II

If ratio of present ages of A and B is $x : y$ and after n yr, the ratio of their ages will be $p : q$, then

$$\frac{x+n}{y+n} = \frac{p}{q}$$

Ex. 2 I am three times as old as my son. 15 yr hence, I will be twice as old as my son. What is the sum of our present ages.

Sol. Let the present age of man and son be $3k$ and k respectively,

So, $x = 3k$ and $y = k$

So, $\frac{p}{q} = \frac{2}{1}$ $\quad [\because 15yr$ hence their ratio is $2:1]$

Then, $\frac{x+n}{y+n} = \frac{p}{q} \implies \frac{3k-15}{k-15} = \frac{2}{1}$

$\implies \quad 3k-15 = 2(k-15) \implies 3k-15 = 2k-30$

$\therefore \qquad k = 30-15 = 15$

$\therefore$ Sum of their present ages $= 3k + k = 4k = 4 \times 15 = 60 \ yr$

Note

Mostly questions on ages can be solved with the use of linear equations. So, the method to solve linear equations is important for this chapter which are discussed in chapter equations.

9.2 Fast Track Techniques to solve the QUESTIONS

Technique 1

If $t \ yr$ after age of one person is n times of that of another person and at present the age of first person is m times of that of another person, then

$$\text{Age of the first person} = tm\left(\frac{n-1}{m-n}\right) yr$$

$$\text{and Age of the second person} = t\left(\frac{n-1}{m-n}\right) yr$$

Ex. 3 Present age of Karan is 5 times the age of Shivam. After 10 yr, Karan will be 3 times as old as Shivam. What are the present ages of Karan and Shivam?

Sol. Let the present age of Shivam be a $x\ yr$. Then, present age of Karan $= 5x\ yr$

After 10 yr, the ratio of ages will be 3 : 1

According to the question,

$$\frac{5x+10}{x+10} = \frac{3}{1}$$

$$\implies 5x+10 = 3(x+10)$$

$$\implies 5x+10 = 3x+30)$$

$$\implies 2x = 20 \implies x = \frac{20}{2} = 10$$

$\therefore$ Karen's present age $= 5 \times 10 = 50\ yr$ and Shivams's present age $= 10\ yr$

Fast Track Method

Here, $t = 10, m = 5$ and $n = 3$

$$\text{Karen's present age} = tm\left(\frac{n-1}{m-n}\right)$$

$$= 10 \times 5\left(\frac{3-1}{5-3}\right) = 50 \times \frac{2}{2} = 50\ yr$$

$$\text{Shivam's present age} = t\left(\frac{n-1}{m-n}\right)$$

$$= 10\left(\frac{3-1}{5-3}\right) = 10 \times \frac{2}{2} = 10\ yr$$

Technique 2

If $t_1\ yr$ before, age of a person was m times of age of another person. After $t_2\ yr$, age of a person will be n times of age of second person, then

$$\text{Age of First person} = \frac{t_2 m(n-1) + t_1 n(m-1)}{m-n}\ yr$$

$$\text{and Age of Second person} = \frac{t_2(n-1) + t_1(m-1)}{m-n}\ yr$$

Ex. 5 yersr ago, age of Sachin was 4 times the age of Vineet and after 10 yr, Sachin will be twice as old as Vineet. Find the present ages of Sachin and Vineet. Sol. Let present age of Sachin and Vineet be xyr and yyr respectively,
Case 1. 5 years ago,

$$\frac{x-5}{y-5} = \frac{4}{1}$$

$$\implies \quad x-5=4(y-5)$$
$$\implies \quad x-5=4y-20$$
$$\implies \quad x-4y=-20+5$$
$$\implies \quad x-4y=-15 \qquad (1)$$

Case 2. After 10 years,

$$\frac{x+10}{y+10}=\frac{2}{1}$$
$$\implies \quad x+10=2(y+10)$$
$$\implies \quad x+10=2y+20$$
$$\implies \quad x-2y=20-10$$
$$\implies \quad x-2y=-10 \qquad (2)$$

On subtracting Eq. (1) from Eq. (2), we get

$$x-4y-(x-2y)=-15-10$$
$$x-4y-x+2y=-25$$
$$-2y=-25 \implies y=\frac{25}{2}=12.5$$

On putting value of y in Eq. (1), we get

$$x-4(\frac{25}{2})=-15$$
$$x-50=-15 \implies x=-15+50=35$$

Hence, present age of Sachin is *35 yr* and present age of Vineet is *12.5 yr*.

Fast Track Method

Here, $t_1=5, t_2=10, m=4$ and $n=2$

$$\text{Present age of Sachin}=\frac{t_2m(n-1)+t_1n(m-1)}{m-n}$$
$$=\frac{10\times4(2-1)+5\times2(4-1)}{4-2}=\frac{40+10\times3}{2}=\frac{70}{2}=35\ yr$$
$$\text{Present age of Vineet}=\frac{t_2(n-1)+t_1(m-1)}{m-n}$$
$$=\frac{10(2-1)+5(4-1)}{4-2}=\frac{10+5\times3}{2}=\frac{25}{2}\ yr$$

Technique 3

If M is as elder to N as he is younger to P and sum of ages of N and P is $t\ yr$, then N<M<P

$$\text{M's Age} = \frac{\text{Sum of age of N and P}}{2} = \frac{S}{2}, \qquad \text{where S=Sum}$$

Ex. 5 If Akshay is as much elder than Vinay as he is younger to Kartik and sum of ages of Vinay and Kartik is 48 yr, then find the age of Akshay.

Sol. Let the present age of Akshay be $x\ yr$ and he is younger to Kartik by $y\ yr$

Then, Kartik's age $= (x+y)\ yr$ and Vinay's age $= (x-y)\ yr$.

Now, according to the question,

Sum of ages of Vinay and Kartik $= 48$

$\Longrightarrow \quad (x+y)+(x-y) = 48$

$\Longrightarrow \quad 2x = 48$

$\therefore \quad x = \frac{48}{2} = 24\ yr$

Hence, Present age of Akshay $= 24\ yr$.

Fast Track Method

$$\text{Present Age of Akshay} = \frac{\text{Sum of ages}}{2} = \frac{48}{2} = 24\ yr$$